

Chapter 7 Study Sheet

Quadrilaterals and Other Polygons — everything you need on one page.

7.1 Angles of Polygons

- Interior sum: $(n - 2) \cdot 180^\circ$. Regular n-gon, each angle: $(n - 2) \cdot 180 \div n$.
- Exterior angles always total 360° . Regular n-gon, each: $360 \div n$. Given each exterior angle e , the polygon has $360 \div e$ sides.
- Vocabulary: equilateral (sides) · equiangular (angles) · regular (both) · convex vs. concave (does a diagonal escape?).

7.2 Properties of Parallelograms

- Opposite sides $\cong$ · opposite angles $\cong$ · consecutive angles supplementary · diagonals bisect each other.
- Coordinates: the fourth vertex is $R = Q + (S - P)$; check that both diagonals share the same midpoint.

7.3 Proving a Quadrilateral Is a Parallelogram

- Five conditions: opposite sides parallel (definition) · opposite sides $\cong$ · opposite angles $\cong$ · ONE pair of opposite sides parallel AND $\cong$ · diagonals bisect each other.
- The classic trap: one pair parallel and the OTHER pair congruent is NOT enough (isosceles trapezoid).
- Coordinate proofs: slopes show parallel; the distance formula shows congruent; midpoints show bisecting diagonals.

7.4 Special Parallelograms

- Rectangle = $\square$ + four right angles; its diagonals are CONGRUENT.
- Rhombus = $\square$ + four congruent sides; its diagonals are PERPENDICULAR and bisect the corner angles.
- Square = rectangle + rhombus at once — it inherits every property in the chapter.
- The diagonals of a square make 45-45-90 triangles: diagonal = side $\cdot \sqrt{2}$.

7.5 Trapezoids and Kites

- Trapezoid: exactly one pair of parallel sides (bases); leg angles are co-interior, so they add to 180° .
- Isosceles trapezoid $\iff$ base angles congruent $\iff$ diagonals congruent.
- Midsegment = $\frac{1}{2}(\text{base}_1 + \text{base}_2)$, parallel to both bases.
- Kite: perpendicular diagonals; exactly one pair of congruent opposite angles (between the non-congruent sides).